

TECHNICAL DOCUMENTATION

Arapoint

Nigerian Identity & Utility Verification Platform

System Architecture · API Reference · Developer Roadmap

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Table of Contents

1.
Tech
Stack
& Arc
hitect
ure [3](#)

2.Dat
abas
e Sc
hem
a [4](#)

3.Mo
dule
1 —
Public
Marketing
Site
[5](#)

4.Mo
dule
2 —
Main
Platf
orm
Dash
boar
d [5](#)

5.Mo
dule
3 —
Deve
loper
API
Portal
[6](#)

6.Em
ploy
ment
Verifi
catio
n En
dpoin
t [8](#)

7.Mo
dule
4 —
Admin
Control

Panel
[9](#)

8.
RPA
Work
er
System

10

9.Security

Model

10

10.S
uggested I
mpro
vements
& Ro
admap 11

1. Tech Stack & Architecture

Frontend

Technology	Role	Notes
React 18 + TypeScript	UI framework	Component-based SPA
Vite	Build tool	Hot module replacement, fast builds
TailwindCSS	Styling	Utility-first CSS framework
shadcn/ui	Component library	Radix UI primitives + Tailwind
Wouter	Client routing	Lightweight React router

Backend

Technology	Role	Notes
Node.js + Express	HTTP server	REST API, port 5000
TypeScript (tsx)	Runtime	No compile step — direct execution
Drizzle ORM	Database layer	Type-safe PostgreSQL queries
PostgreSQL	Database	Replit-managed hosted instance
bcrypt	Security	Password + secret key hashing
jsonwebtoken	Auth	JWT for developer portal sessions

External Integrations

Service	Purpose	Auth Method
Prembly (IdentityPass)	NIN + BVN verification	PREMBLY_SECRET_KEY env var
Replit Object Storage	File/document storage	Managed integration
OpenAI	AI features (available)	Managed integration

Two Concurrent Workflows

The platform runs two Node.js processes simultaneously:

- Server Backend — Express API server on port 5000, handles all HTTP traffic
- RPA Worker — Standalone job processor that polls the database queue and executes browser automation for async verification tasks (WAEC, NECO, IPE, etc.)

Repository Structure

```
Arapoint/
% % % client/                React SPA (all frontend code)
%   % % % src/pages/         Marketing site pages
%       % % % (public)
%       % % % dashboard/     Main platform UI
%       % % % developer/     B2B Developer portal (dark theme)
%       % % % admin/         Admin control panel
% % % server/
%   % % % src/
%       % % % api/routes/     Express route handlers
%       % % % services/       External API integrations (Prembly, etc.)
%       % % % db/             Drizzle ORM schema & DB connection
%       % % % rpa-worker.ts    Standalone RPA job processor
% % % exports/               Generated files (PDFs, reports)
```

2. Database Schema

Core Developer Tables

Table	Purpose	Key Columns
developer_users	Developer accounts	id, email, password_hash, wallet_balance, kyc_status
developer_api_keys	API credential pairs	api_key, secret_key_hash, secret_key_last_four, environment
developer_api_logs	API call audit trail	developer_id, endpoint, status_code, cost_duration_ms
developer_transactions	Wallet history	developer_id, transaction_type, amount, description

RPA & Service Tables (shared with main platform)

Table	Purpose	Key Columns
rpa_jobs	Async verification queue	service_type, query_data, status, result, error_message
kyc_documents (column)	KYB structured data	JSONB on developer_users — companyInfo, directors, apiUseCase,

All developer tables use inline Drizzle ORM definitions in developer.ts. An idempotent migration (ALTER TABLE ... ADD COLUMN IF NOT EXISTS) runs on server startup to ensure new columns exist without downtime.

3. Module 1 — Public Marketing Site

Fully built React pages with teal/green brand theme. No authentication required.

Route	Page	Notes
/	Home	Hero, features overview, CTA
/features	Features	Platform capability showcase
/services	Services	NIN, BVN, Employment Verification, CAC
/pricing	Pricing	Tiered plans — no NGN amounts shown publicly
/about	About	Company information
/careers	Careers	Job listings
/contact	Contact	Enquiry form
/privacy	Privacy Policy	NDPR-aligned privacy policy
/terms	Terms of Service	Legal terms
/developers	Developer Hub	API overview landing page

4. Module 2 — Main Platform Dashboard

Internal/enterprise verification dashboard. Requires staff login. Services available:

Service	Provider	Mode
NIN Lookup	Prembly	Synchronous
BVN Verification	Prembly	Synchronous (basic + premium face match)
WAEC/SSCE Result	RPA Worker	Asynchronous — queued job
NECO Result	RPA Worker	Asynchronous — queued job
NABTEB Result	RPA Worker	Asynchronous — queued job
CAC Business Lookup	External API	Synchronous
IPE / IPPIS Clearance	RPA Worker	Asynchronous — queued job
NIN Personalization	RPA Worker	Asynchronous — queued job

5. Module 3 — Developer API Portal (B2B)

Developer Onboarding Flow

1. Developer visits /developer/login !' clicks "Create Account"
2. Enters email !' receives OTP (via otpService)
3. Submits OTP + name + password !' account created
4. Server auto-creates Sandbox keypair !' credentials returned ONCE
5. Developer logs in !' receives 7-day JWT for portal session
6. Developer uses API Key (X-API-Key header) for all API calls

Three Developer Credentials

Credential	Format Example	Purpose
Account ID	a3f2c1d0-...UUID...	Unique account identifier — for support & billing
API Key	ara_sand_48f3a... / ara_live_c9d2b...	Authenticates API calls via X-API-Key header
Secret Key	ara_sk_sand_... / ara_sk_live_...	Signs webhook payloads from Arapoint to developer

Security: Secret keys are hashed with bcrypt (cost factor 10) before storage. Only the last 4 characters are stored in plain text. The raw secret key is revealed exactly once — immediately after creation — and cannot be recovered.

Sandbox !' Live Mode Progression

```
Register !' Sandbox Mode (auto sandbox keypair created)
!"
Developer tests API with sandbox keys (ara_sand_xxx)
!"
Developer submits KYB (5-step business verification form)
!"
Admin reviews KYB !' Approve / Conditional / Reject
!"
On Approval !' Admin promotes developer to Live Mode
!"
Developer can now create Live keys (ara_live_xxx)
Live keys !' process real identity data
```

Developer Portal Pages

Page	URL	Function
Login / Register	/developer/login	OTP-verified account creation
Dashboard	/developer/dashboard	Stats, recent calls, wallet balance
API Credentials	/developer/api-keys	Account ID, sandbox/live keys + secret keys
API Logs	/developer/logs	All call history with cost and status
Billing	/developer/billing	Wallet balance, transactions, top-up
Documentation	/developer/docs	API reference with code examples
Business Verification	/developer/kyb	5-step KYB form (KYC for businesses)
Account Settings	/developer/account	Profile, password change, KYB status

API Endpoints (all require X-API-Key header)

Endpoint	Method	Cost (₦)	Description
/verify/nin	POST	130	NIN lookup via Prembly — name, DOB, photo, address
/verify/bvn	POST	80	BVN lookup via Prembly — name, DOB, phone, bank
/verify/education	POST	250	SSCE/WAEC/NECO/ NABTEP — queried via RPA
/verify/education/result	GET	Free	Poll RPA job status by jobId
/verify/unified	POST	400	Combined NIN + BVN + Education in one call
/verify/employment	POST	350–450	Full employment check — see Section 6

Example API Request

```
curl -X POST https://arapoint.com.ng/api/v1/developer/verify/nin \
-H "X-API-Key: ara_sand_your_api_key_here" \
-H "Content-Type: application/json" \
-d '{"nin": "12345678901"}'
```

Response:

```
{
  "status": "success",
  "code": 200,
  "message": "NIN verification completed",
  "data": {
    "verification": {
      "success": true,
      "data": {
        "firstName": "John",
        "lastName": "Doe",
        "dateOfBirth": "1990-01-15",
        "gender": "Male",
        "phone": "080XXXXXXX",
        "address": "12 Sample Street, Lagos"
      }
    }
  }
}
```


6. Employment Verification Endpoint

How It Works — System Flow

```
POST /api/v1/developer/verify/employment
Body: { nin, bvn, ssce: { provider, examYear, registrationNumber }, level }

Step 1: Validate inputs (NIN 11 digits, BVN 11 digits, SSCE provider valid)
Step 2: Deduct developer wallet ( |350 standard / |450 higher)
Step 3: Call Prembly API for NIN  % % % % % % % % % % % % % % % % % % % %
Step 4: Call Prembly API for BVN  % % % % % % % % % % % % % % % % % % % %$ parallel
Step 5: Queue RPA job for SSCE  % % % % % % % % % % % % % % % % % % % %
Step 6: Cross-reference – name match + DOB match (NIN != BVN)
Step 7: Compute weighted confidence score
Step 8: Return full checkpoint result immediately
      (SSCE result arrives later – caller polls jobId)
```

Confidence Scoring System

Checkpoint	Weight	Notes
NIN Verified	35 pts	Full name, DOB, address confirmed via NIMC
BVN Verified	30 pts	Financial identity confirmed via CBN/ bank record
Name Cross-Match (NIN != BVN)	10 pts	Fuzzy match — handles order differences
DOB Cross-Match (NIN != BVN)	10 pts	Date of birth consistency check
SSCE Qualification Verified	15 pts	Async — added when RPA job completes
TOTAL	100 pts	If SSCE not provided, max scales to 85 pts

Confidence Grade Thresholds

Grade	Score Range	Label
A	90 – 100%	Very High Confidence
B	75 – 89%	High Confidence
C	55 – 74%	Moderate Confidence
D	35 – 54%	Low Confidence
F	0 – 34%	Very Low Confidence

Example Response Structure

```
{
  "status": "success",
  "data": {
    "requestId": "EMP-A3F2C1D0B4E5...",
    "level": "standard",
    "confidence": {
      "score": 85,
      "label": "High Confidence",
      "grade": "B",
      "earned": 85,
      "maxPossible": 100
    },
    "checkpoints": {
      "nin": { "status": "verified", "earned": 35, "data": { ... } },
```

```
"bvn": { "status": "verified", "earned": 30, "data": { ... } },
"crossMatch": {
  "nameMatch": { "result": true, "earned": 10 },
  "dobMatch": { "result": true, "earned": 10 }
},
"ssce": {
  "status": "processing",
  "jobId": "uuid-of-rpa-job",
  "pollUrl": "GET /verify/education/result?jobId=...",
  "maxScore": 15, "earnedScore": 0
}
}
}
```

7. Module 4 — Admin Control Panel

Separate authentication layer (admin session). Full-access management interface.

Admin Sections

Section	Capability
Platform Stats	Total developers, active count, total API calls, platform revenue, pending KYB
Developer List	Searchable table — all developers with mode, KYB status, wallet, active/inactive status
Developer Detail Panel	Slide-in panel: full profile, API keys, usage summary, recent logs, action buttons
API Logs	All platform calls — developer name, endpoint, method, status, cost, duration
KYB Review Queue	Structured KYB data view — 3-way decision (Approve / Conditional / Reject)

Admin API Endpoints

Endpoint	Method	Function
/admin/developers	GET	Paginated developer list
/admin/developers/:id	GET	Full detail: profile + keys + logs + usage summary
/admin/developers/:id/status	PATCH	Activate or deactivate account
/admin/developers/:id/promote	PATCH	Promote to Live mode (requires approved KYB) or move to Sandbox
/admin/stats	GET	Platform-wide statistics
/admin/logs/all	GET	All API logs with developer name/email (joined query)
/admin/kyc	GET	KYB queue filtered by status
/admin/kyc/:id	PATCH	Submit KYB decision (approve/conditional/reject)

KYB (Business Verification) Form — 5 Steps

Step	Data Collected
1 — Company Info	Legal name, CAC number, business type, date of incorporation, address, state, TIN, phone, website
2 — Directors / UBO	All directors: full name, DOB, nationality, ID type + number, ownership percentage
3 — API Use Case	Purpose, expected monthly volume, target customers, revenue model, specific APIs needed
4 — Compliance	PEP declaration, AML declaration, data usage agreement, developer terms acceptance
5 — Review & Submit	Collapsible summary of all sections before final submission

8. RPA Worker System

A separate Node.js process (rpa-worker.ts) that runs continuously alongside the API server.

RPA Worker lifecycle:

1. Polls rpa_jobs table WHERE status = 'pending' (interval: ~30s)
2. Claims job !' sets status = 'processing'
3. Executes automation against target portal (WAEC, NECO, IPPIS, etc.)
4. Parses response !' stores result JSON in rpa_jobs.result
5. Sets status = 'completed' (or 'failed' with error_message)
6. If developer has webhook_url !' sends signed POST with result payload

Supported Job Types

Service Type	Target Source	Output
waec_result	WAEC portal	Subject results, grades, year, registration number
neco_result	NECO portal	Subject results, grades, year
nabteb_result	NABTEB portal	Technical qualification results
nbais_result	NBAIS portal	Islamic education board results
jamb_score	JAMB portal	UTME scores, subject breakdown
ipe_clearance	IPPIS portal	Payroll/civil service clearance status
nin_personalization	NIMC portal	NIN slip generation and retrieval

9. Security Model

Authentication Layers

Layer	Mechanism	Scope
Developer portal session	JWT (HS256, 7-day expiry)	Portal UI routes (profile, keys, logs, billing)
API calls	API Key in X-API-Key header	All /verify/* endpoints
Admin panel	Admin JWT + adminAuth middleware	All /admin/* endpoints
Webhook verification	HMAC-SHA256 using Secret Key	Incoming webhook to developer server

Data Protection

- Passwords hashed with bcrypt (cost factor 10)
- Secret keys hashed with bcrypt — only last 4 characters stored in plain text
- API keys stored in plain text (needed for lookup) — rotatable on demand
- NIN and BVN numbers masked in API logs (first 4 + *** stored)
- KYB documents stored as encrypted JSONB — never returned in list endpoints
- Admin endpoints require separate credential — completely isolated from developer auth

10. Suggested Improvements & Roadmap

High Priority

Feature	Description	Complexity
Webhook Delivery System	Send signed POSTs to developer webhook URLs when async jobs complete	Medium
Real Payment Integration	Replace mock wallet funding with Paystack or Flutterwave for actual NGN	Medium
Email Notification System	KYB status changes, low wallet balance alerts, job completion emails via SendGrid	Low
Rate Limiting	Per-API-key request limits: sandbox (100/day), live (based on plan); return 429 with Retry-After	Medium

Medium Priority

Feature	Description	Complexity
NYSC Verification	Complete Employment Higher tier — add NYSC check via BDA to the verified endpoint	Medium
Sandbox Mock Responses	Return realistic simulated data in sandbox mode so developers can build their integrations	Low
Developer SDK	Official Node.js and Python client libraries with full TypeScript types and examples	Medium
API Versioning	Move to /api/v2/ with structured deprecation policy and breaking changes	Medium
Team Access	Allow developers to invite team members with role-based permissions (admin/read-only)	High

Lower Priority

Feature	Description	Complexity
Admin Audit Log	Track every admin action: who approved which KYB, who promoted to live, who deleted a record	Low
Usage Alerts	Developer wallet low-balance alerts via webhook/email at configurable thresholds	Low
Analytics Dashboard	Developer-facing charts: calls per day, endpoint breakdown, success rate	Medium
IP Allowlisting	Restrict API key usage to specific IP ranges (per key configuration)	Low
Batch Verification API	Accept arrays of records for bulk NIN/BVN/employment checks with async responses	High

The platform foundation is solid. The most impactful next steps are: (1) webhook delivery so developers get real-time async results, (2) Paystack payment integration to enable real wallet funding, and (3) sandbox mock responses to dramatically improve developer onboarding experience.